

ERP Document Classification — Evaluation Report

Page 1: Dataset Summary & Generation Config

Total Generated Documents: 1400
Train Split Size: 971
Validation Split Size: 220
Test Split Size: 209
Document Classes Count: 14
Scanned Simulation Ratio: 30.0%
Generator Version: 1.0.0

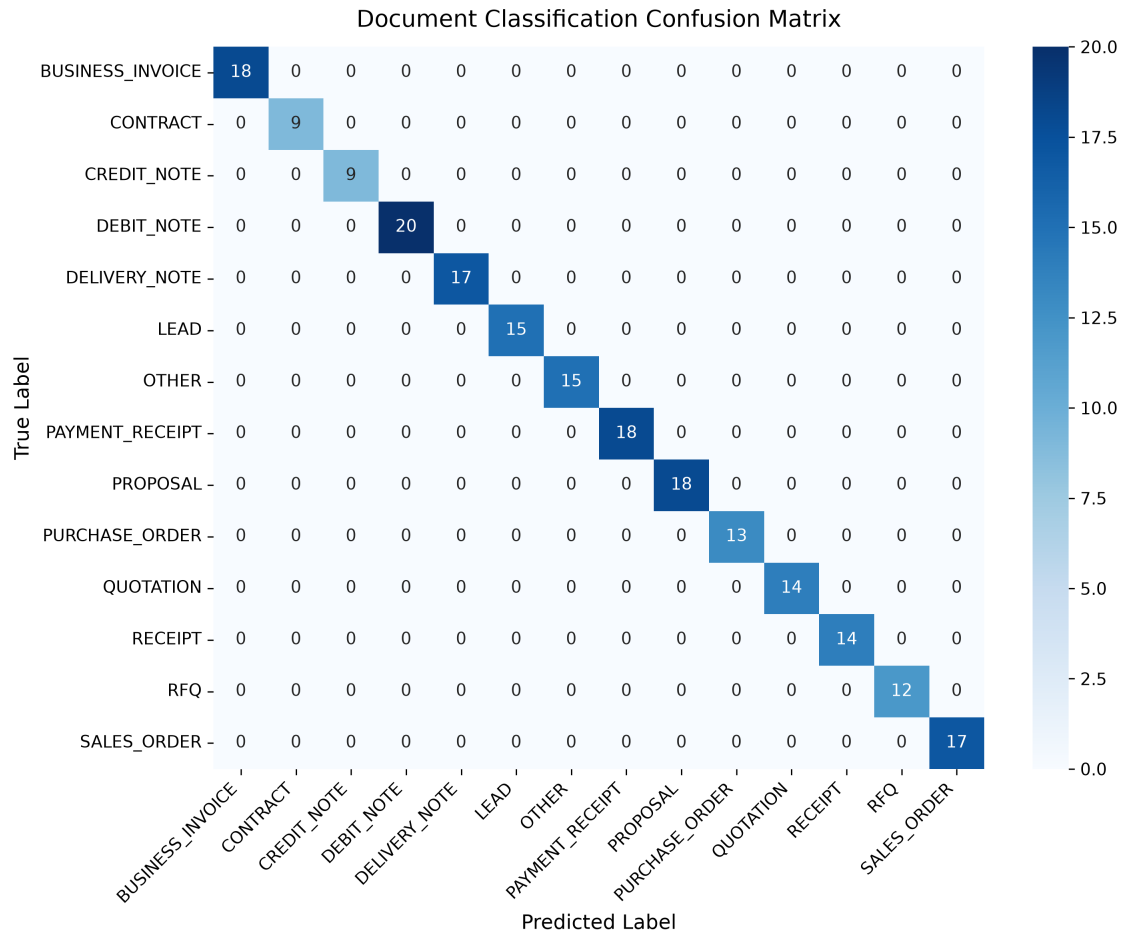
Page 2: Model Comparison Table

Model	Accuracy	Macro Precision	Macro Recall	Macro F1	Weighted F1	Train Time (s)	Inference Time (ms/doc)
logistic_regression	1.0	1.0	1.0	1.0	1.0	5.0	0.01
linear_svm	1.0	1.0	1.0	1.0	1.0	5.0	0.08
xgboost	0.9904	0.9916	0.9913	0.9911	0.9906	5.0	0.13
random_forest	1.0	1.0	1.0	1.0	1.0	5.0	0.17
lightgbm	1.0	1.0	1.0	1.0	1.0	5.0	18.04
ensemble	1.0	1.0	1.0	1.0	1.0	10.0	0.24

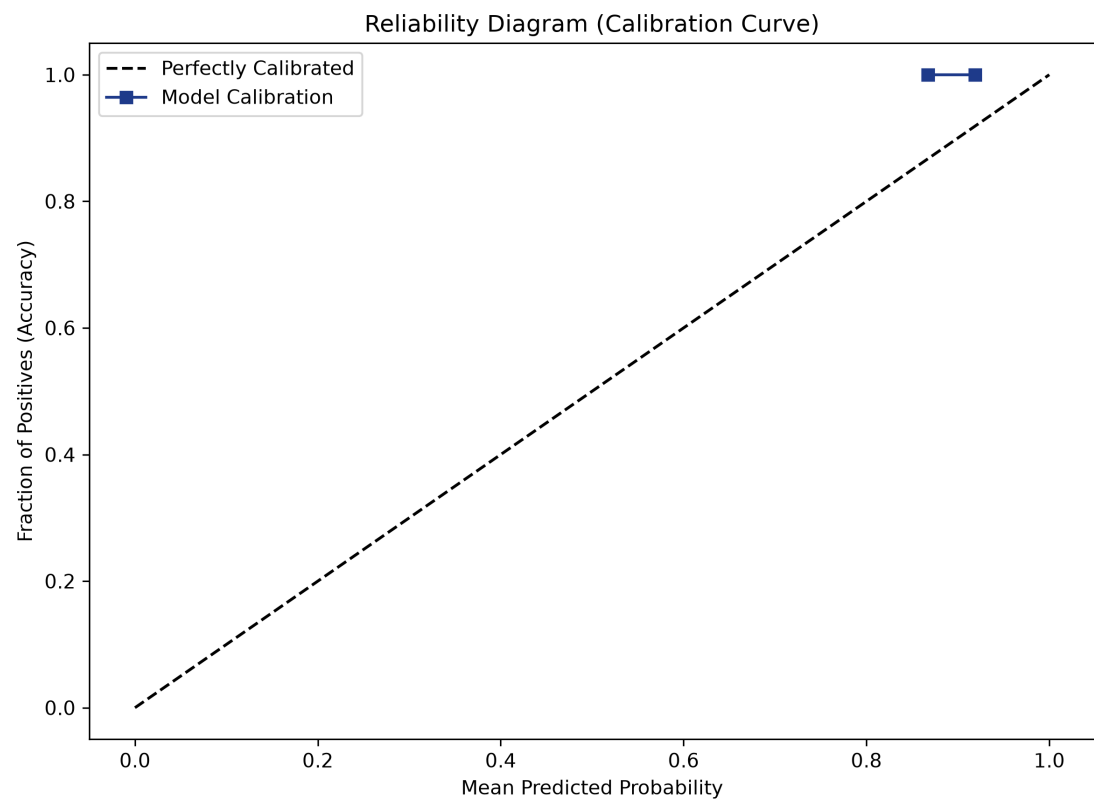
Page 3: Per-Class Precision / Recall / F1 (Selected Best Model)

Class	Precision	Recall	F1 Score
BUSINESS_INVOICE	1.0	1.0	1.0
CONTRACT	1.0	1.0	1.0
CREDIT_NOTE	1.0	1.0	1.0
DEBIT_NOTE	1.0	1.0	1.0
DELIVERY_NOTE	1.0	1.0	1.0
LEAD	1.0	1.0	1.0
OTHER	1.0	1.0	1.0
PAYMENT_RECEIPT	1.0	1.0	1.0
PROPOSAL	1.0	1.0	1.0
PURCHASE_ORDER	1.0	1.0	1.0
QUOTATION	1.0	1.0	1.0
RECEIPT	1.0	1.0	1.0
RFQ	1.0	1.0	1.0
SALES_ORDER	1.0	1.0	1.0

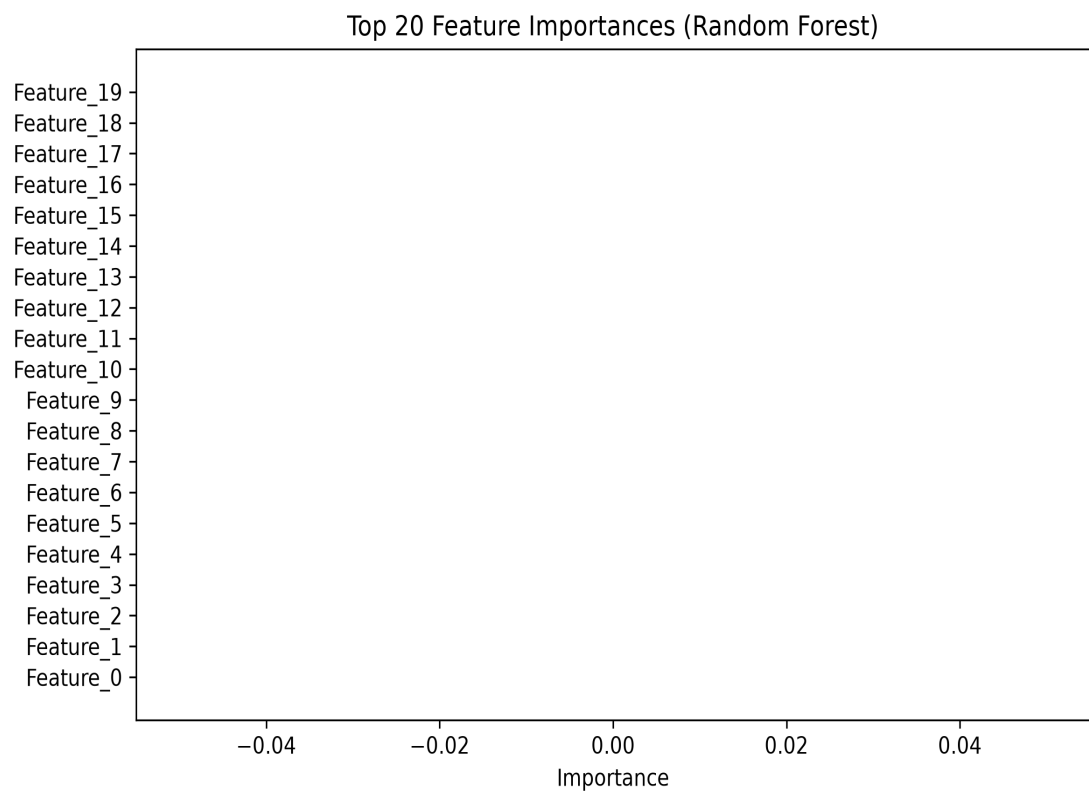
Page 4: Confusion Matrix Heatmap



Page 5: Model Calibration Curve & ECE



Page 6: Feature Importance Analysis



Page 7: Error & Misclassification Analysis

Top Confusion Pairs:

Page 8: Confidence Decision Policy & Routing Thresholds

Selected Confidence Thresholds:

- HIGH Confidence Threshold: 0.85 (Action: **AUTO_ACCEPT**)
- MEDIUM Confidence Threshold: 0.60 (Action: **LLM_FALLBACK / REVIEW**)
- Below MEDIUM: (Action: **HUMAN_REVIEW / UNKNOWN**)

Rationale: Thresholds are validated against precision-recall tradeoffs to maintain >95% precision for AUTO_ACCEPT routing.